

Performance Optimization of the AI4Land Training Pipeline on MareNostrum 5

Ferran Mirabent Rubinat

Master's Thesis in Modelling for Science and Engineering
Universitat Autònoma de Barcelona

Tutors:

Oriol Tintó Prims
Barcelona Supercomputing Center (BSC)

Sílvia Cuadrado Gavilán

Universitat Autònoma de Barcelona (UAB)

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Abstract

Accurate modelling of land use and land cover (LULC) change is essential for projecting carbon emissions and understanding climate feedbacks. The AI4Land project, developed at the Barcelona Supercomputing Center within the framework of the Horizon Europe projects CONCERTO and TerraDT, employs a UNet-based deep learning model to downscale LULC data from the Land-Use Harmonization dataset (LUH2) at approximately 30 km resolution to 1 km resolution, covering the period 1850–2100 including future projections based on Shared Socioeconomic Pathways (SSPs). The pipeline is also intended to be adopted by the ELLIOT project for training geospatial foundation models. Training this model on multi-channel geospatial datasets is computationally demanding and requires efficient use of high-performance computing resources.

This thesis focuses on identifying and resolving performance bottlenecks in the AI4Land training pipeline running on MareNostrum 5, equipped with NVIDIA H100 GPUs. Preliminary profiling has revealed the data loading stage as the primary bottleneck, with significant variance in epoch duration traceable to the current use of multiple Zarr stores and suboptimal data ingestion patterns. The work will systematically evaluate optimizations with a particular emphasis on the data loading pipeline, while also exploring computational improvements at the model and hardware level. Each optimization will be assessed through profiling with NVIDIA Nsight Systems to quantify its impact on throughput and training time.

The expected outcome is a measurably faster and more efficient training pipeline that maintains model accuracy, enabling faster iteration cycles for the AI4Land project and contributing practical guidelines for training deep learning models on large-scale HPC infrastructure.